

From Multimodal LLM to Human-level AI: Modality, Instruction, Reasoning and Beyond

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ABSTRACT

Artificial intelligence (AI) encompasses knowledge acquisition and real-world grounding across various modalities, including language, visual, auditory, and sensory data. Multimodal large language models (MLLMs) have thus recently garnered growing interest in both academia and industry, showing an unprecedented trend to achieve human-level AI. This tutorial aims to deliver a comprehensive review of cutting-edge research in MLLMs, focusing on three key areas: MLLM architecture design, instructional learning, and multimodal reasoning of MLLMs. We will explore technical advancements, synthesize key challenges, and discuss potential avenues for future research. All the resources and materials will be made available online: <https://mllm2024.github.io/ACM-MM2024>

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1 INTRODUCTION

This year, the whole world has witnessed astonishing advancements in AI to date due to the emergence of LLMs, such as OpenAI's ChatGPT [25] and GPT-4 [24]. LLMs have showcased remarkable capabilities in understanding language, hinting at the not-so-distant arrival of true AGI. In reality, we humans inhabit a world where various modalities of information coexist, including visual, auditory, sensory and more, beyond pure language. This realization

underscores the necessity of endowing LLMs with multimodal perception and comprehension capabilities. This endeavor has given rise to an emerging topic of Multimodal LLMs. Researchers devise additional encoders in front of textual LLMs for receiving inputs in other modalities, leading to the development of MLLMs, such as BLIP-2 [13], Flamingo [1], MiniGPT-4 [44], Video-LLaMA [37], LLaVA [21], PandaGPT [27], SpeechGPT [36] and NExT-GPT [31].

As the manner of interactions with LLMs has shifted into a more human-centric 'query-answer' style, the learning of LLMs has also changed. Different from the typical training of deep models, e.g., masked language modeling [3], instruction tuning has been introduced as a major approach for LLMs/MLLMs' tuning [27, 34]. With sufficient instruction learning, LLMs/MLLMs are taught to faithfully follow human instructions. Also, it is critical to fully exploit the potential of LLMs/MLLMs for achieving human-level reasoning. Researchers have designed the Chain-of-Thought (CoT) concept [29], which offers a solution enabling LLMs with complex problem-solving abilities on language [5, 28] or multimodal data [40, 42]. Unfortunately, while technical advancement has been rapidly updated in this topic, a comprehensive and instructional introduction and tutorial to this field can be lacking. In this tutorial, we aim to offer holistic instruction that contributes to achieving stronger, more intelligent, and more human-level AI. We will delve into recent progress in the realm of MLLMs under three major parts, **multi-modality architecture design, instruction learning, multimodal reasoning**.

2 TUTORIAL OUTLINE

2.1 Part I: Introduction and Overview

We begin motivating the MLLMs with the current progress in both academia and industry for achieving the goal of human-level AI. And then we place the emphasis on the key aspects of building successful MLLMs, which bring out the following tutorial content.

2.2 Part II: Architecture Design and Evolution

We delve into the development of MLLMs based on the success of textual LLMs. We review the architecture design and training

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techniques of existing popular MLLMs from three main aspects. (1) We summarize vanilla MLLM architectures that integrate LLMs with different modality information [2, 13, 21], including multi-modal encoding, fusion and generation [31, 35]. (2) We review the pretraining techniques to learn foundational MLLM capabilities from large-scale multimodal data [2, 8, 11, 26]. (3) We introduce the latest advancements in MLLM: from instance-level multimodal input encoding and multimodal generation to fine-grained levels (e.g., pixel-level visual generation) [9, 14, 30, 38].

2.3 Part III: Multimodal Instruction Tuning

Recent instruction-tuned MLLMs including [1, 17, 19, 31, 37, 44] have shown remarkable performance. In this part, we delve into how to build instruction-tuned MLLMs step by step. (1) We introduce the construction of visual instruction data and the approaches to improve data quality and scale up. (2) We engage in the intricate details of the training strategies, like MiniGPT4 [44], LLaVA [21], etc. (3) We discuss some challenges, including parameter-efficient training and relieving hallucination issues [16, 18].

2.4 Part IV: Multimodal Reasoning

Enabling LLMs to reason like humans is also the key to realizing human-level AI, a milestone that has been a focal point of artificial intelligence research for decades. Multimodal reasoning is a rapidly evolving research field that aims to enable them to learn from information gathered from various sources and engage in complex reasoning [2, 4, 12, 22, 32, 33]. We will delve into the techniques and methodologies of multimodal reasoning. (1) We introduce benchmark and evaluations. (2) We explore key research topics, including multimodal chain-of-thought reasoning [42], multimodal in-context learning [43], and compositional reasoning [23]. (3) We discuss the challenges and future research directions, including multimodal tool learning and multimodal autonomous agents.

2.5 Part V: Panel Discussion

All the presenters will gather around, and meanwhile we will invite other external researchers, which all together comprise the panelists. This will be a starting point for an in-depth discussion on how current MLLM systems are likely to evolve and develop, and how to model a world that can ultimately lead to our final goal.

3 SPECIFICATION OF THE TUTORIAL

Presentation Format. We prefer a **half-day** 4-hour tutorial. We hope the tutorial will have good knowledge coverage while not taking the audiences too much additional time. Our tutorial covers three major aspects, each involving specific and varied content.

Target Audience and Prerequisites. This tutorial is targeted at members of a broad range of relevant machine learning communities, e.g., NLP, CV and broad AI. This includes researchers, students of both academia and industry, as well as practitioners wishing to make use of LLMs. Based on the level of interest in this topic, we expect around **200–300** participants.

Materials, Reading List and Resources. The tutorial will make all teaching material available online, including slides, reading list, representative tools, data resources, code resources and pre-record videos openly available at <https://mllm2024.github.io/ACM-MM2024>.

4 TUTORIAL PRESENTERS

Hao Fei (<https://haofei.vip>). He is currently a Research Fellow in the School of Computing, National University of Singapore; and also an Associate Researcher at Skywork AI, Kunlun 2050 Research, Singapore. His research interests lie in LLM and multimodal learning. Over 50 of his research papers have been published at top-tier venues. He was honored as a Rising Star at the 2023 World Artificial Intelligence Conference Yunfan Award. His representative work includes NExT-GPT [31], Vitron [9], Video-of-Thought [7], Dysen-VDM [6] and Finsta [10].

Xiangtai Li (<https://lxtgh.github.io/>). is a research scientist at TikTok, Singapore. He worked as a Research Fellow in S-Lab, and a member of the Multimedia Laboratory of NTU (MMLab@NTU) at Nanyang Technological University in 2023. His research interests include computer vision and machine learning with a focus on scene understanding, segmentation, video understanding, and multi-modal learning. Several of his works have been published in top-tier venues.

Haotian Liu (<https://hliu.cc/>). He is a Ph.D. student at the University of Wisconsin–Madison. His research interest lies in LLMs, and recent focus is on building customizable large models that follow humans' intent. He co-organized the ECCV 2022 and CVPR 2023 workshop on Computer Vision in the Wild. His representative work includes Gligen [15], LLaVA [21], LLaVA-1.5 [20].

Fuxiao Liu (<https://fuxiaoliu.github.io>). He is currently a Ph.D. student in the school of Computer Science, University of Maryland, College Park. His recent focus is on building customizable large models that follow humans' intent. His research has been published at top-tier venues. His representative work includes HallusionBench [16], LRV-Instruction [17], MMC [19] and Visual News [17].

Zhuosheng Zhang (<https://bcmi.sjtu.edu.cn/~zhangzs/>). He is an Assistant Professor at Shanghai Jiao Tong University, China. His research interests include NLP, LLMs, and MLLM agents. He was honored as a Rising Star at the 2023 World Artificial Intelligence Conference Yunfan Award and won outstanding doctoral dissertation award of CIPS. His representative work includes UMR [39], Auto-CoT [41], MM-CoT [42], and Auto-GUI [40].

Hanwang Zhang (<https://personal.ntu.edu.sg/hanwangzhang/>). He is an Assistant Professor at Nanyang Technological University, Singapore. His research interests include computer vision, multimedia, and social media. Dr. Zhang is the recipient of the Best Demo runner-up award in ACM MM 2012, the Best Student Paper award in ACM MM 2013, and the Best Paper Honorable Mention in ACM SIGIR 2016, and TOMM best paper award 2018. He is the winner of Best Ph.D. Thesis Award of National University of Singapore, 2014.

Shuicheng Yan (<https://yanshuicheng.info/>). Prof. Yan is managing director of Kunlun 2050 Research and Co-CEO of Skywork AI, and former Group Chief Scientist of Sea Group. Prof. Yan Shuicheng is a Fellow of Singapore's Academy of Engineering, AAIL, ACM, IEEE, and IAPR. His research areas include computer vision, machine learning, and multimedia analysis. His team won more than ten best papers and best student paper awards, particularly a grand slam at the ACM Multimedia, the top-tiered conference in multimedia, including the Best Paper Awards thrice, Best Student Paper Awards twice, and Best Demo Award once.

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